

■ The Dangers of Backtesting

This book has shown you quite a few backtest results and has drawn plenty of conclusions from them, so it may seem odd that I now warn you about backtesting. As a tool, backtesting can be very helpful but as the expression goes, a little bit of knowledge can be a dangerous thing. This book is not about backtesting, mainly because that's a subject which would require not only a book on its own, but multiple books to properly explain it. But I will try to give you the gist of the problem, to make you aware of the main pitfall.

There is an old industry adage: "I have never seen a backtest that I didn't like", meaning that you can, on purpose or by accident, make a backtest look exactly the way you want it to. After all, you are analysing static, historical data and you can keep tweaking until the results show the kind of numbers you were looking for. The way most people approach backtesting is the so-called *Groundhog Day* method, or infinite trial and error. For the slightly younger readers, perhaps *The Edge of Tomorrow* may be easier to relate to. Perhaps you start out with a theory about how you can apply some indicator to a market, and then you go test it. After it didn't work out very well, you study the details and find that your stop loss point was a little too tight, so you move it a little further away. Your results improve, but not enough, so you go study the details some more, and decide to add a trend filter. And this iterative process can go on and on, until you learn to become a better human being like Bill Murray or kill some giant blue alien like Tom Cruise. But in this case, you continue until you have a perfect-looking backtest, with amazing results. This is likely to lull you into a false sense of security, and make you think that you have somehow discovered the Holy Grail of trading.

It's really quite easy to make a perfect backtest, one which appears to prove the possibility of a Sharpe ratio of 15, of hundreds of percent per year return with almost no drawdowns. But such backtests are merely curve fitting and lack any sort of predictive power, and with a devastatingly high probability, they will crash and burn in real life.

Since this really isn't a book about backtesting, I would very much encourage you to pick up a copy of the excellent book *Systematic Trading*, by Robert Carver (2015), which is the best book I am aware of on this particular topic.